Standard Operating Procedure: Biomedical Waste Management and Spill Handling

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Department	Laboratory Safety	Status	Approved
Prepared by	Quality Manager	Reviewed by	Consultant Pathologist
Approved by	Laboratory Director	Distribution	All collection and processing staff

1. Purpose

To segregate, store and dispose of biomedical waste safely, and to respond correctly to blood spills and needle-stick injuries, in line with the Bio-Medical Waste Management Rules, 2016 (as amended).

2. Scope

Applies to all laboratory, collection and housekeeping staff at every BIODNA site.

3. Responsibilities

Role	Responsibility
All staff	Segregate waste at the point of generation into the correct colour-coded container
Housekeeping	Collect, weigh and move bags to the central storage area; maintain the waste register
Safety officer	Monthly audit, staff training, liaison with the authorised waste treatment facility

4. Waste segregation by colour code

Container	What goes in	Lab examples
Yellow bag	Soiled waste and pre-treated laboratory waste	Blood-soaked cotton and gauze; microbiology cultures and plates after autoclaving; blood bags
Red bag	Contaminated recyclable plastic	Gloves, empty sample tubes, urine containers, tubing, syringes without needles
White translucent puncture-proof container	Sharps	Needles, needles fixed to syringes, lancets, scalpel blades
Blue cardboard box	Glassware	Broken or used glass slides, glass vials, glass tubes
Black bag (general waste)	Non-infectious waste only	Paper, packaging, food waste. Never biomedical waste

Microbiology cultures and stocks must be pre-treated (autoclaved) on site before they go into the yellow bag.

5. Handling sharps

- Never recap, bend or break needles by hand.
- Drop used needles into the white sharps container immediately at the point of use.
- Close and replace the sharps container when it is three-quarters full. Never overfill or push waste down.

6. Storage and disposal

- Tie bags when three-quarters full and label them with the date, department and BIODNA site.
- Move waste to the locked central storage area at least once a day.

- Waste must be handed over to the authorised common treatment facility within 48 hours of generation.
- Record the weight of each category in the biomedical waste register daily.

7. Blood or body fluid spill

1. Warn others and restrict access to the area.
2. Wear gloves, a lab coat and eye protection. For large spills, also wear a disposable apron.
3. If there is broken glass, pick it up with forceps, never with hands, and place it in the blue box.
4. Cover the spill with absorbent paper or a spill kit pad.
5. Pour 1% sodium hypochlorite (freshly prepared) gently over the paper, working from the outside in.
6. Leave for a contact time of 10-20 minutes.
7. Wipe up and discard all material in the yellow bag. Clean the area again with detergent and water.
8. Record the spill in the incident register.

8. Needle-stick or sharps injury

1. Do not panic and do not squeeze or suck the wound. Let it bleed freely for a few seconds.
2. Wash immediately with running water and soap for several minutes. Do not use bleach on skin.
3. For a splash to the eyes or mouth, rinse with plenty of clean water.
4. Report to the section supervisor immediately and fill in the incident report form.
5. Go to the designated medical officer at once. Post-exposure prophylaxis (PEP), if required, works best when started within 2 hours and must start within 72 hours.
6. Identify the source patient if possible; the medical officer arranges source testing with consent.

9. Records

Biomedical waste register (BIODNA-F-SAF-001), incident register (BIODNA-F-SAF-002), staff training records. Retain for 5 years.

10. References

- Bio-Medical Waste Management Rules, 2016, Ministry of Environment, Forest and Climate Change, Government of India (as amended).
- ISO 15190: Medical laboratories: Requirements for safety.
- NABL 112: Specific criteria for accreditation of medical laboratories.

11. Revision history

Version	Date	Change	Approved by
2.0	2026-04-01	Added needle-stick PEP timing and 48-hour disposal rule	Laboratory Director
1.0	2024-04-01	First issue	Laboratory Director